Calculus

Objects

1) Explain the concept of calculus

2) Define differential calculus

3) Examine the limits, continuity and derivatives of a function

4) Define integral calculus

5) List down the differential and integral formulas

What is calculus?

Calculus is the study of change. It

provides a framework for modeling systems in which there is change and ways to make predictions about such models

- Calculus is one of the key areas of maths that deals with continuous change

- It is based on 2 key concepts: derivatives and integrals.

- It is also referred to as Infinitesimal calculus

- The amount that are virtually equal to nothing but are still not quite zero are known as infinitesimal numbers.

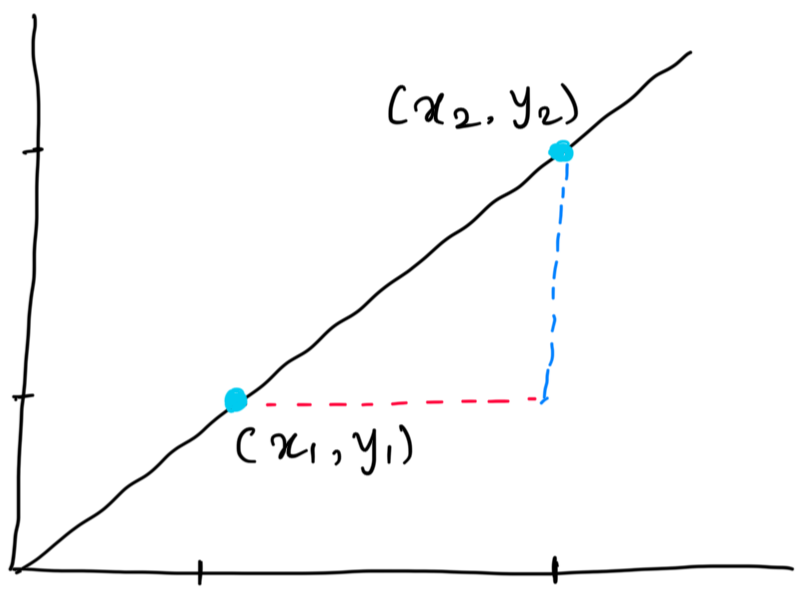
- By a large, old-style calculus is the investigation of continuous change of function.

Derivatives	Integrals
The derivative of a function is a proportion of the pace of progress of a function, which helps in finding the slope of curve	The integral of a function is a proportion of the area under the curve of maths function $F(x)$ plotted as a function of X.
It explains the function at a specific point.	It collects the discrete benefits of a function over

	a range of values.

Differential Calculus

It is a part of calculus that deals with the study of the rates at which quantities change



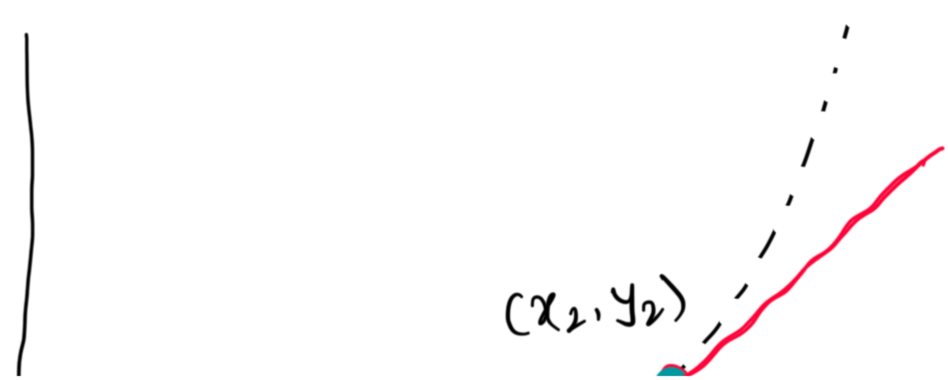
- Consider a scenario where x and y be two real numbers such that y is a function of x, that is, $y = f(x)$

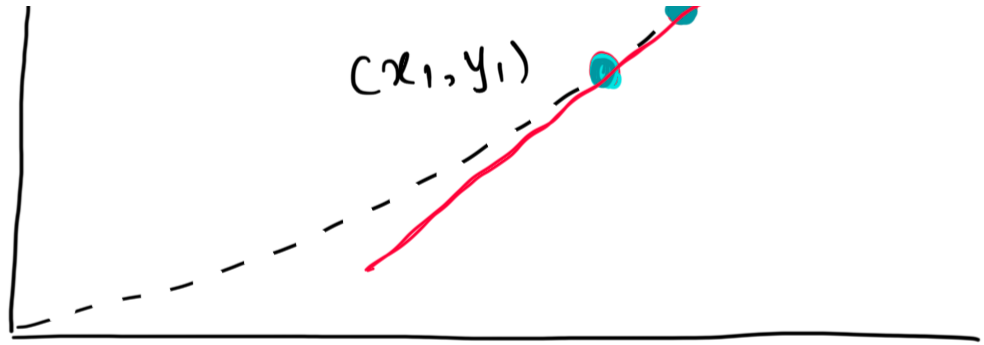
- If $F(x)$ is the equation of straight line (linear equation.) then the equation is represented as $y = mx + b$, where m represents the slope

The following equation determines the value of m in the slope

Differential Calculus equation

$$\boxed{m = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x}}$$





The $\Delta y/\Delta x$, also represented as dy/dx, is the derivative of y w.r.t. x, as well as the rate of change of y per unit change in X.

The slope of curvature the graph, which varies at various points, represents the slope of an imaginary straight line drawn through that small graph segment.

Limits

In maths, limits are defined as the values that a function approaches as the input approaches a certain value.

Limit equation

$$\boxed{\lim_{x \to c} f(x) = A}$$

This expression is read as
"The limit of F of x as x approaches C equals A."

Derivative

It is defined as the varying rate of c' unge of ı Function w.r.t. an independent variable.

Derivative Equation

$$\boxed{\lim_{x \to h} [f(x+h) - f(x)]/h = A}$$

Derivatives are used to measure the sensitivity of one dependent variable w.r.t. another independent variable.

Continuity

A function $f(x)$ is assumed to be continuous at a specific point where $x = a$, if and only if its three circumstances are fulfilled.

1) $f(a)$ is defined
2) $\lim_{x \to a} f(x)$ exists
3) $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a)$

Differential Formulas

Differential formulas can be applied to basic algebraic expressions, trigonometric ratios, inverse trigonometry and exponential terms.

$$\frac{d}{dx} x^n = n x^{n-1}$$

$$\frac{d}{dx} a^x = a^x \log a$$

$$\frac{d}{dx} \text{constant} = 0$$

$$\frac{d}{dx} \log x = \frac{1}{x}$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

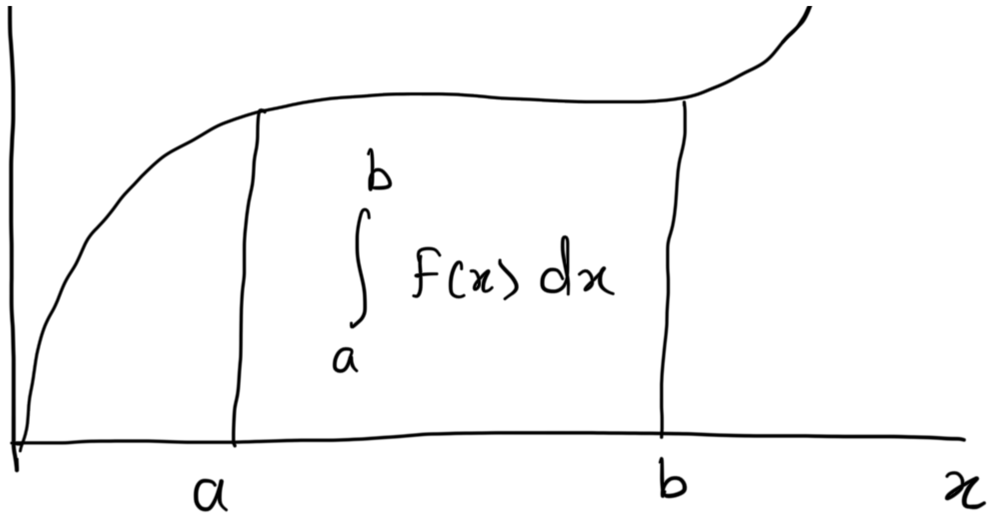
$$\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$$

$$\frac{d}{dx} \sec x = \sec x \tan x$$

$$\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x$$

Integral Calculus

It assigns a number to a functions to describe displacement, area, volume and other concepts that arise by combining infinitesimal data.



Consider a function f of a real variable x and an interval $[a, b]$ of the real line

The definite integral is defined as the signed area of the region in xy-plane that is bounded by the graph of f, the x-axis and vertical lines $x = a$ and $x = b$

An integral is the inverse of a differential & vice versa.

Integral calculus

$$\boxed{\int_a^b f(x)\,dx}$$

Definite Integral

- A definite integral has specific limit or cutoff for the computation of the function

The upper & lower limits of a free factor of a function are highlighted

Definite integral Equation

$\int^b$

$$\int_a f(x)\,dx = F(b) - F(a)$$

Indefinite Integral

An indefinite integral does not have a particular limit, for example, no upper & lower limit is characterized

The integration value is then constantly connected by a constant value (C).

Indefinite Integral Equation

$$\boxed{\int f(x)\,dx = F(x) + C}$$

Integration Formulas

Integrals formulas can be derived

From differentiation formulas, and are complimentary to differentiation formulas.

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + c \qquad \int 1 \, dx = x + c$$

$$\int e^x \, dx = e^x + c \qquad \int \left(\frac{1}{x}\right) dx = \log|x| + c$$

$$\int a^x \, dx = \frac{a^x}{\log a} + c \qquad \int \cos x \, dx = \sin x + c$$

$$\int \sin x \, dx = -\cos x + c$$

$$\int \sec^2 x \, dx = \tan x + c$$

$$\int \mathrm{cosec}^2 x \, dx = -\cot x + c$$

$$\int \sec x \tan x \, dx = \sec x + c$$

$$\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + c$$

Key Takeaways

- Differential calculus is a part of the calculus that deals with the study of the rates at which quantities change

- Integral calculus assigns numbers to functions to describe displacement, area, volume and other concepts that arise by combining infinitesimal data

- A definite integral has a particular

limit or cutoff for the computation of the function

- Derivatives address the momentary pace of progress of an amount w.r.t. another.